THE PROBLEM

Farm information should be *spoken*, not searched.

- Scheme rules are scattered: PM-Kisan exclusions, Krishak Unnati registration, paddy procurement tokens, crop insurance deadlines.
- Crop advice must match the crop's stage, and pesticide doses must come from trusted sources.
- Weather forecasts don't say *"should I irrigate today?"*
- Most portals are text-heavy and in English; many farmers prefer to **speak in Hindi or Hinglish**.

> 🎤 “मैं 2 एकड़ का किसान हूँ, इनकम टैक्स नहीं भरता, KCC नहीं है” → 🔊 which schemes fit, why, and what to ask next.

5	5 + 1	2	4–7 s
schemes + eligibility rules	crop guides + monthly calendar	languages: हिंदी · Hinglish	voice question → spoken answer

THE SOLUTION · THREE SAATHIS IN ONE VOICE APP

📋 Yojana Saathi

Answers questions on PM-Kisan, Krishak Unnati / dhaan kharidi, PM Fasal Bima, KCC and Soil Health Card, and checks eligibility from what the farmer says, with an editable form.

🌱 Crop Saathi

Stage-wise advice for dhaan, chana, gehun, teora and makka, plus "what to sow this month". Doses only from ICAR / NCIPM, with a KVK reminder.

🌦️ Mausam Saathi

7-day district forecast with rule-based advice: skip irrigation before rain, frost and heat alerts, disease-weather warnings.



Speak or type · हिंदी | Hinglish toggle · Answer cards + charts · Web sources when the KB has no answer

Male voice · Timbre Yashraj

Anish Deep Toppo · B.Tech Data Science & AI (final year) · IIIT Naya Raipur

Live: **kisan-saathi.duckdns.org** Demo: **youtu.be/phamRC-49Hc**

01 / SPEAK
Browser mic
Farmer taps the mic and asks; the recording (≤ 25 s) goes over HTTPS to FastAPI and ffmpeg makes 16 kHz WAV.

→

02 / LISTEN
Gnani Prisma v3.0
Speech to text with **indic-auto**; handles Hindi-English mixed speech, returns the detected language.

→

03 / ROUTE
Keywords → Gemini
Clear questions are routed by keywords (crop, stage, district); unclear ones go to Gemini 3.1 Flash-Lite as JSON.

→

04 / ANSWER
Grounded reply
Uses only retrieved knowledge base text or live data, in the selected language; returns an answer card.

→

05 / SPEAK BACK
Gnani Timbre v3.0
Male voice **Yashraj** reads the reply (Devanagari or Roman Hinglish); v2.5 Deepak as fallback.

ROUTER → MODULE → SOURCE

Yojana schemes	RAG over 5 source-checked scheme guides → scheme card with documents, how to apply, helpline
Eligibility "main 2 acre..."	Profile pulled from speech → rule engine (fail / needs check / eligible) → editable form, re-check needs no LLM call
Crop pests · stage	RAG filtered by crop & stage over ICAR / NCIPM / IIWBR / IIMR guides + monthly calendar → crop card, translated, doses unchanged
Mausam weather	Open-Meteo 7-day forecast + advisory rules (rain > 10 mm/48 h, frost < 8°C, heat > 35°C) → Chart.js card

WHEN THE KNOWLEDGE BASE HAS NO ANSWER

KB score < 0.70 → SearXNG (self-hosted) → DuckDuckGo → Summary with sources → Labelled general answer, no doses

BUILT WITH

Gnani Prisma v3.0 · Gnani Timbre v3.0 · Gemini 3.1 / 3.5 Flash-Lite · gemini-embedding-2 · FastAPI · Python · NumPy vector index · ffmpeg · SearXNG + ddgs · Open-Meteo · HTML / CSS / JS · Chart.js · AWS Lightsail · Caddy HTTPS · Docker

SPEED

- Keyword fast path skips the router call on most questions
- Fast Flash-Lite model first, backup model after a timeout
- Voice question → spoken answer in about **4–7 s**

TRUST & SAFETY

- Answers only from retrieved text; numbers and doses copied exactly
- Pesticide doses only from official ICAR / NCIPM guides, with source link
- Every crop answer ends with "confirm with your KVK / RAEO"
- Web answers are labelled and list their sources; general answers with doses or ₹ amounts are blocked
- Keys in `.env` only; per-IP rate limit; 139 unit tests